Rakesh Surampalli

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Professional Summary

Full Stack and AI Engineer with 5+ years of experience building production-grade web applications, AI agents, LLM-powered workflows, backend APIs, databases, integrations, and customer-facing product experiences. Strong hands-on experience with TypeScript, JavaScript, React, Next.js concepts, Python, Django, REST APIs, PostgreSQL, SQL, OpenAI APIs, Gemini concepts, Anthropic Claude workflows, RAG, prompt engineering, AI agents, tool calling, workflow automation, and cloud deployment practices. Experienced taking ambiguous problems from concept to prototype to production by working directly with users, translating real-world workflows into product requirements, building frontend and backend features, integrating external systems, testing with users, and iterating based on feedback. Startup builder with experience building FinAI and Tudu from scratch across architecture, product discovery, full stack development, AI integration, deployment, testing, and production support. Open to San Francisco relocation and in-person collaboration.

Core Technical Skills

Frontend: TypeScript, JavaScript, React.js, Next.js Concepts, HTML5, CSS3, Responsive UI, Component-Based Development, Product UX, Dashboard Interfaces

Backend: Python, Django, FastAPI Concepts, REST APIs, Webhooks, API Integrations, Microservices Concepts, JSON, Authentication, Authorization

AI and LLMs: AI Agents, LLM Applications, OpenAI API, Gemini Concepts, Anthropic Claude, Prompt Engineering, RAG, Tool Calling, Function Calling, Context Management

AI Product Engineering: Agentic Workflows, AI-Powered Applications, Prototype-to-Production Delivery, Model Evaluation, Response Validation, Guardrails, AI-Assisted Development

Databases and Data: PostgreSQL, SQL Server, MySQL, MongoDB Concepts, SQL Queries, Data Modeling, Data Pipelines, Query Debugging, Analytics Workflows

Integrations: ERP Concepts, CRM Concepts, Third-Party APIs, Google Maps API, Market Data APIs, Internal Tools, Data Services, Workflow Automation

Cloud and DevOps: AWS Concepts, GCP Concepts, Azure Concepts, Docker, Git, GitHub Actions, CI/CD, Environment Configuration, Deployment Validation

Product and Collaboration: Customer Discovery, User Testing, Product Requirements, Zero-to-One Product Building, Startup Execution, Technical Communication, Agile Delivery

Professional Experience

Anatham Intelligence **Mar 2026 to Present**

Forward Deployed Engineer and Full Stack AI Engineer *United States*

- Build and ship full stack AI product features across Tudu and FinAI using TypeScript, JavaScript, React, Python, Django, PostgreSQL, REST APIs, OpenAI APIs, LLM workflows, and agentic automation patterns.
- Own ambiguous product problems from concept to prototype to production by gathering requirements, designing technical workflows, building frontend and backend features, testing with users, deploying changes, and iterating based on feedback.
- Develop end-to-end AI agents for planning, recommendation generation, financial insight generation, task automation, summarization, user support, and workflow execution across production product use cases.
- Design AI workflows that combine user intent, business rules, retrieved context, tool calls, API responses, structured outputs, validation checks, and user feedback loops into reliable product experiences.
- Build React-based product experiences including dashboards, onboarding screens, task planning views, portfolio analysis screens, shared workflows, responsive layouts, and AI-assisted user interfaces.
- Develop backend APIs, database models, service logic, integration workflows, and data pipelines to connect AI agents with user records, product dashboards, recommendation history, and third-party systems.
- Prototype new AI product features quickly, validate them with real users, collect feedback, identify failure modes, and improve usability, reliability, and product-market fit through rapid iteration.
- Integrate external APIs and services including OpenAI APIs, Google Maps APIs, market data sources, authentication workflows, internal backend services, and data systems to support production use cases.
- Apply RAG and retrieval patterns using embeddings, semantic search, context construction, metadata filtering, prompt engineering, and response validation to improve accuracy of LLM-generated outputs.
- Evaluate and integrate new AI capabilities by testing prompt patterns, comparing model behavior, reviewing output quality, validating response structure, and selecting the right model behavior for each use case.
- Use AI-assisted development tools including ChatGPT, Claude, Cursor concepts, Codex-style workflows, and GitHub Copilot to accelerate straightforward engineering tasks while reviewing code quality and correctness.
- Support infrastructure and production readiness through Git, Docker, CI/CD concepts, environment configuration, deployment checks, SQL validation, API testing, logging, and post-release verification.
- Debug issues across frontend, backend, APIs, databases, authentication, deployment configuration, and AI workflows by reviewing logs, SQL records, API responses, browser behavior, and model outputs.
- Work directly with users and stakeholders to understand operational workflows, translate business needs into product requirements, explain technical tradeoffs, and deliver clear product improvements.

– Create technical documentation, product notes, API guides, prompt behavior notes, troubleshooting guides, implementation plans, and customer-facing onboarding materials.

CVS Health **Jun 2022 to Feb 2023**

Software Engineer *United States*

– Developed enterprise web application features using Python, JavaScript, Java, Spring Boot, REST APIs, SQL, PostgreSQL, cloud-based workflows, and internal data systems.

– Built backend API services to process application metadata, monitoring outputs, risk indicators, business rules, structured records, and reporting datasets for internal stakeholders.

– Collaborated with product, security, platform, business, and operations teams to translate requirements into scalable technical solutions, validate workflows, and deliver maintainable application features.

– Developed RESTful services following request validation, business logic execution, database persistence, structured JSON response handling, exception handling, and API testing practices.

– Supported secure enterprise workflows involving application security, compliance-oriented reporting, internal process automation, sensitive data handling, monitoring outputs, and operational dashboards.

– Implemented AI-assisted knowledge discovery workflows using LLM concepts, prompt engineering, semantic search, structured retrieval, and RAG-style patterns for internal document intelligence use cases.

– Created SQL queries using joins, filters, aggregates, metadata lookups, and reporting logic to support analytics workflows, compliance review, issue investigation, and operational visibility.

– Supported testing, debugging, release validation, defect resolution, and production troubleshooting using Postman, Swagger/OpenAPI, application logs, database checks, and root cause analysis.

– Created technical documentation for APIs, data flows, system behavior, troubleshooting steps, security considerations, implementation details, and support workflows.

Copart **Jun 2021 to May 2022**

Software Engineer *Dallas, TX*

– Built full stack and data-driven application workflows using Python, JavaScript, React concepts, Java, REST APIs, SQL, PostgreSQL, SQL Server, Tableau, Power BI, and cloud-based development practices.

– Partnered with business, operations, analytics, and engineering stakeholders to understand workflow gaps, define requirements, automate manual analysis, and deliver scalable technical solutions.

– Developed API-driven workflows and reusable backend logic to support operational reporting, asset tracking, valuation analysis, workflow visibility, data validation, and business intelligence requirements.

– Created SQL queries using joins, aggregates, filters, grouping, ranking logic, and date-based analysis to identify delayed workflows, data quality issues, inventory aging, and operational bottlenecks.

– Built dashboards and analytical views to communicate insights clearly to business users, helping teams understand performance patterns, workflow exceptions, throughput, and process gaps.

– Supported testing, validation, defect investigation, and production issue resolution by tracing issues across SQL logic, API behavior, dashboard outputs, source records, and business rules.

– Improved operations by converting recurring manual reporting tasks into reusable queries, dashboard logic, backend data processing steps, and documented troubleshooting procedures.

– Documented data definitions, reporting logic, technical workflows, recurring issue patterns, and support procedures to improve maintainability and team knowledge sharing.

Toyama Controls and Systems **May 2019 to Jan 2021**

Back End Developer *India*

– Developed backend services and automation workflows using Python, JavaScript, SQL, REST APIs, MVC-style patterns, and database modules for IoT-enabled operational systems.

– Built data processing workflows to capture customer usage data, automation events, sensor signals, device status updates, connectivity events, and system performance information.

– Supported integration patterns connecting backend services, device data, operational dashboards, user actions, and automation logic for real-time workflow execution.

– Troubleshot application and system issues by reviewing logs, API outputs, database records, user reports, device behavior, and edge-case scenarios.

– Created technical documentation for backend workflows, database behavior, issue investigation steps, deployment notes, and operational support procedures.

Selected Product Work

Tudu: AI Smart Task Planner Built from Scratch

TypeScript, React, Next.js Concepts, Python, Django, PostgreSQL, REST APIs, OpenAI API, Google Maps API, AI Agents

– Built and launched a production AI application from scratch that converts natural language task inputs into structured tasks, dates, priorities, group workflows, recommendations, and location-aware action plans.

– Owned product discovery, technical architecture, frontend workflows, backend APIs, AI features, third-party integrations, deployment validation, testing, and customer-facing support.

– Designed end-to-end AI agents for task interpretation, smart scheduling, duplicate task prevention, user preference handling, location-aware recommendations, and multi-step workflow automation.

– Developed React and TypeScript-based user interfaces for task creation, weekly planning, shared task workflows, protected user flows, onboarding screens, responsive layouts, and API-driven updates.

- Built backend APIs for authentication, task creation, task updates, group member management, shared task acceptance, recommendation history, user context, and AI-generated planning outputs.
- Integrated OpenAI APIs and Google Maps APIs to classify user intent, extract dates, route tasks, recommend nearby stores, and generate user-friendly planning suggestions.
- Improved production reliability through API testing, UI debugging, SQL validation, deployment checks, error handling, feedback review, and iterative product improvements.

FinAI: AI Financial Intelligence Platform Built from Scratch

TypeScript, React, Python, Django, PostgreSQL, REST APIs, OpenAI API, RAG, Embeddings, AI Agents

- Built a production AI platform from scratch for portfolio analysis, stock trend detection, benchmark comparison, risk explanation, news summarization, sentiment analysis, and AI-driven recommendations.
- Designed architecture, backend APIs, database models, frontend workflows, AI retrieval layers, deployment workflows, testing processes, and product feedback loops in a startup-style environment.
- Implemented RAG pipelines combining portfolio data, market context, structured metrics, news signals, sentiment data, embeddings, semantic retrieval, and prompt construction for explainable financial insights.
- Developed backend APIs and PostgreSQL models to manage users, portfolios, financial metrics, watchlists, stock records, AI-generated outputs, dashboard datasets, and recommendation history.
- Created React dashboards to help users understand portfolio performance, stock trends, benchmark comparisons, risk indicators, sentiment scores, and AI-generated explanations.
- Built evaluation workflows to review generated insights, compare prompt behavior, inspect retrieved context, validate response structure, and reduce inaccurate or unsupported AI outputs.
- Improved production readiness through API testing, SQL validation, error handling, logging concepts, deployment checks, prompt evaluation, and user feedback review.

Manufacturing AI Agent and ERP/CRM Integration Prototype

TypeScript, React, Next.js Concepts, Python, REST APIs, PostgreSQL, OpenAI API, Gemini Concepts, Anthropic Claude, ERP/CRM Concepts

- Designed a manufacturing AI agent prototype for quoting, pricing, inventory management, procurement, and scheduling workflows using LLM-powered reasoning, structured data retrieval, and API orchestration.
- Modeled ERP and CRM integration flows involving customer records, quotes, product catalogs, inventory levels, vendor records, purchase requests, pricing rules, and scheduling constraints.
- Built API orchestration patterns that connect user requests, business rules, database lookups, external system calls, AI-generated recommendations, and validated structured outputs.
- Compared OpenAI, Gemini, and Anthropic-style model behavior for different workflow needs including structured extraction, summarization, reasoning, recommendation generation, and user-facing explanations.
- Added reliability patterns including input validation, error handling, fallback responses, response structure validation, logging concepts, human review paths, and production-readiness checks.
- Created product notes explaining how sales teams, operators, and plant managers could use AI agents to reduce manual work, improve quoting speed, and coordinate operational workflows.

Education

The University of Texas at Dallas **2023**
Master of Science in Business Analytics *Richardson, TX*
Manipal Academy of Higher Education **2020**
Bachelor of Science in Mechatronics Engineering *India*

Certifications

- DeepLearning.AI Generative AI with Large Language Models
- DeepLearning.AI LangChain for LLM Application Development
- Google Cloud Professional Machine Learning Engineer